

Order-up-to Inventory Model

Instructor: Sairam Sriraman

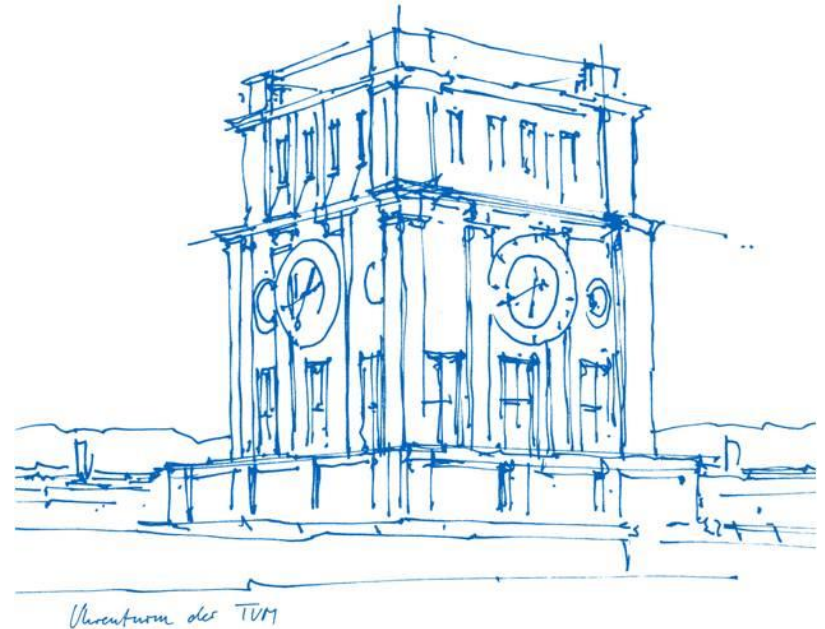
Examiner: Prof. Dr. David Wuttke

Technische Universität München

School of Management

Campus Heilbronn

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What is to criticize about the Newsvendor Model?

Medtronic Case: Supply Chain

This entire session is closely based on Cachon & Terwiesch Chapter 14.

Inventory Management for Medtronic Pacemakers

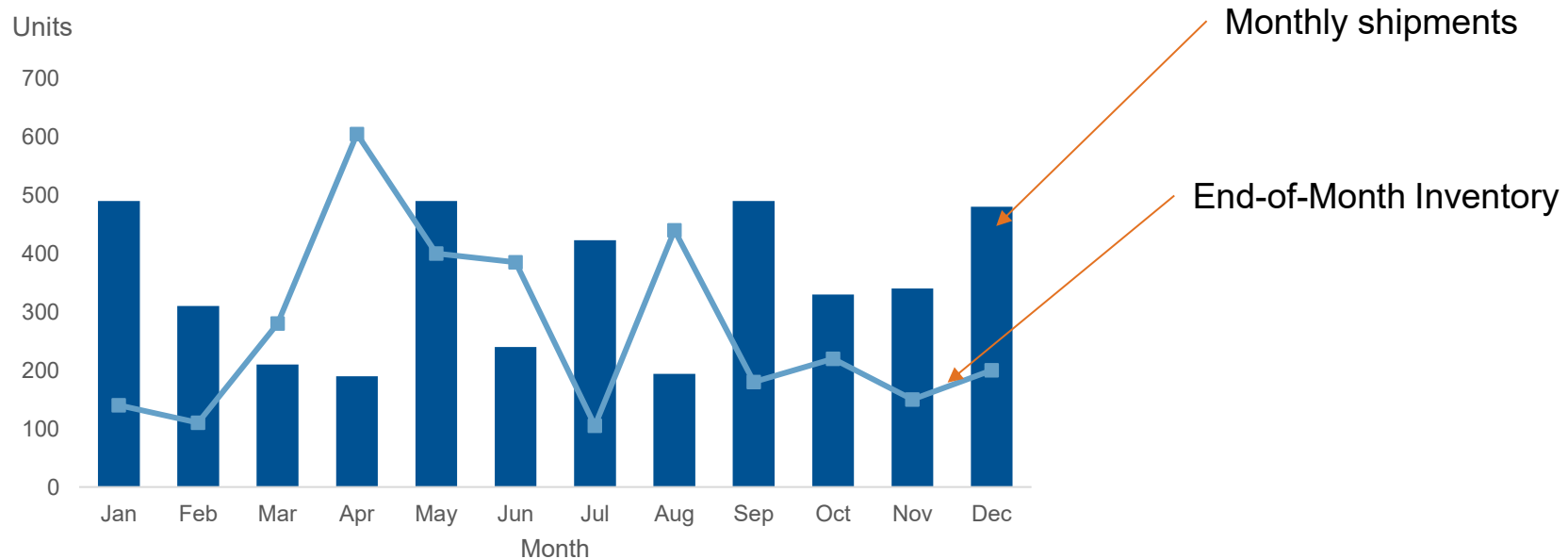


Source of image: <http://newsroom.medtronic.com/phoenix.zhtml?c=251324&p=irol-mediakit&ID=Pacemakers>

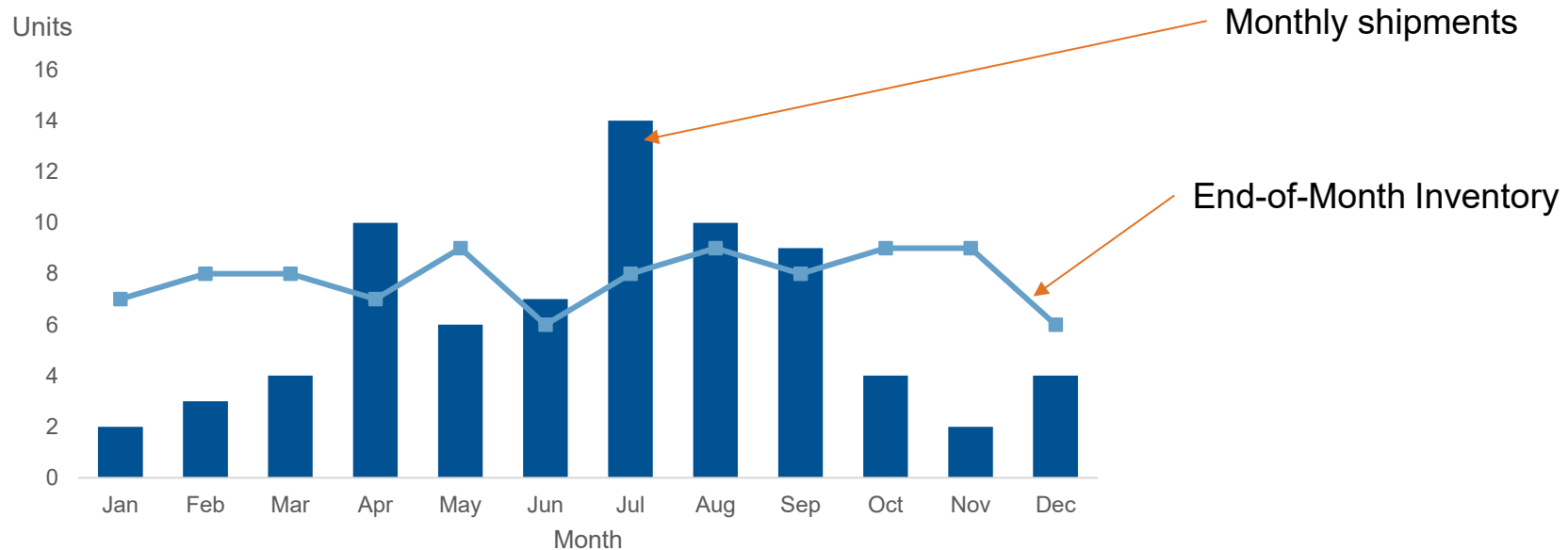
Three Levels for Holding Inventory at Medtronic

- Manufacturing facilities (further ignored since no significant levels of finished goods)
- Distribution Centers (DCs)
- Sales representatives (about 500)

Distribution Center in Mounds View, MN

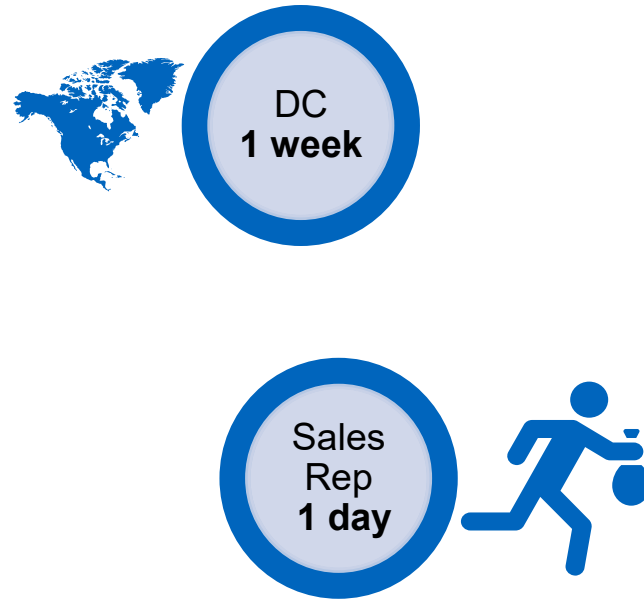


Sales Representative (Susan Magnotto)



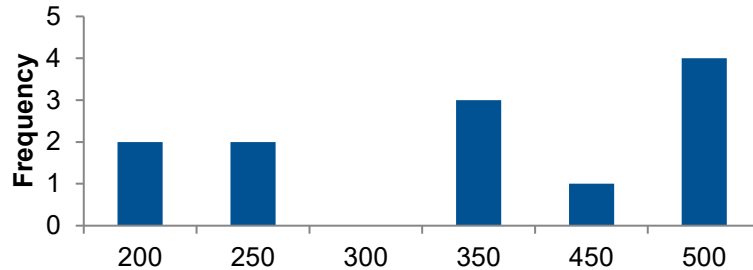
Demand Distributions

Meaningful Periods for Inventory Planning



Distribution Center Level (Mounds View)

Histogram

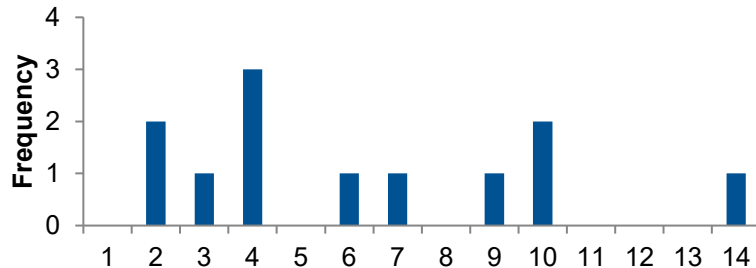


1 period = 1 week

- Mean = 349
 - Standard Deviation = 122.38
 - **Normal Distribution Approximation**
- Mean per week = $349 \cdot \frac{12}{52} = 80.6$
 - Std. dev. per week = $122.38 \cdot \sqrt{12/52} = 58.81$

Sales Representative Level (Susan)

Histogram



- Mean = 6.25
- Standard Deviation = $\sqrt{6.25} = 2.5$
- **Poisson Distribution Approximation**

1 period = 1 day

Assume 5 days per week

- Mean per day = $6.25 \cdot \frac{12}{52} / 5 = 0.29$
- Std. dev. per day = $2.5 \cdot \sqrt{\frac{12}{52 \cdot 5}} = 0.539$

Summary of Demand Distribution Analysis

At this stage...

- You have information about the expected demand distribution. Typically we consider parametric distributions and few parameters are enough.
- Normal distribution is entirely characterized by mean and standard deviation.
- Poisson distribution is entirely characterized by mean.

Instead of demand distribution analysis...

- You could have also carried out a forecast which yields a demand distribution

Order lead times

Order lead time for Distribution Centers = 3 weeks

Order lead time for Sales Representatives = 1 day

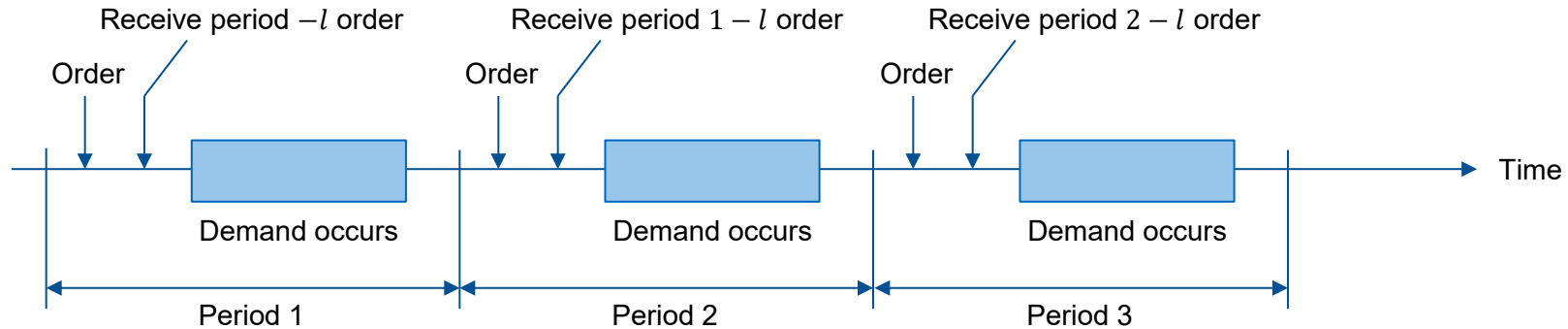
The Order-up-to Model

Concept of Order-up-to Model: Always Order up-to the Same Amount S

- Exactly replaces what has been demanded
- Pull-system (!)
- Potentially several outstanding orders (e.g., one from each week).

Sample Sequence of Events

Order lead time is denoted by l .



Inventory Measures and Variables

Let S denote the order-up-to level.

$$\text{Inventory level} = \text{On-hand inventory} - \text{Back order}$$

$$\begin{aligned}\text{Inventory position} &= \text{On-order inventory} + \text{On-hand inventory} - \text{Back order} \\ &= \text{On-order inventory} + \text{Inventory level}\end{aligned}$$

$$\text{Each period's order quantity} = S - \text{Inventory position}$$

End-of-Period Inventory Level

The **inventory level** measured at the end of a period
equals
the order-up-to level S minus demand over $1 + l$ periods.

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Intuition

- inventory level = inventory position – on-order inventory
- immediately after placing an order: inventory position = order-up-to level S
- on-order inventory = demand over $1 + l$ periods (since you always order the amount that was demanded)

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- inventory level = $\text{inventory position} - \text{on-order inventory}$
- immediately after placing an order: inventory position = order-up-to level S
- on-order inventory = $\text{demand over } 1 + l \text{ periods}$ (since you always order the amount that was demanded)

Stockout Probability

Inventory level at the end of the period = S - demand over $l + 1$ periods

$$\begin{aligned}\text{Stockout probability} &= 1 - \text{Prob}(0 < \text{Inventory level at the end of the period}) \\ &= 1 - \text{Prob}(\text{demand over } l + 1 \text{ periods} < S)\end{aligned}$$


$$:= F(S), \text{ where } F(S) \text{ is the cumulative distribution function of demand over } l + 1 \text{ periods.}$$



If you know S , $F(S)$ gives you the in-stock probability
and $1 - F(S)$ gives you the stockout probability.

Expectations/Expected Values

Expected back orders at the end of any period

$$= \mathbb{E}[\max(0, -(S - \text{Demand over } l + 1 \text{ periods}))] =: B(S)$$

Expected left-over inventory at the end of any period

$$= \mathbb{E}[\max(0, (S - \text{Demand over } l + 1 \text{ periods}))] =: I(S)$$

If you know S , $B(S)$ gives you the expected number of units in back-order and $I(S)$ gives you the expected number of units in inventory.

Expected left over inventory - Expected back orders

$$= I(S) - B(S)$$

$$= \mathbb{E}[\max(0, (S - \text{Demand over } l + 1 \text{ periods}))] - \mathbb{E}[\max(0, -(S - \text{Demand over } l + 1 \text{ periods}))]$$

$$= \mathbb{E}[S - \text{Demand over } l + 1 \text{ periods}] = S - \mu$$

$$\Rightarrow I(S) = S - \mu + B(S)$$

μ is the expected demand.

It suffices to calculate either $B(S)$ or $I(S)$.

Order-up-to Level (S) and Expected Sales $E(z)$

Let Demand be normally distributed with parameters μ and σ . Let $z = \frac{S-\mu}{\sigma}$, where S is the order-up-to quantity. Further, let ϕ denote the density of a standard normal distribution and let Φ denote the corresponding cumulative distribution function.

Observe that $\phi'(z) = \frac{d}{dz} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) = -z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) = -z\phi(z)$. Expected lost sales can be expressed as $L(z) := \int_z^\infty (x - z)\phi(x)dx = -\int_z^\infty -x\phi(x)dx - z \int_z^\infty \phi(x)dx = \phi(z) - z(1 - \Phi(z))$.

With that, expected sales, $E(z) := \mu - \sigma L(z)$ with $L(z) = \phi(z) - z(1 - \Phi(z))$.



It suffices to remember this formula.
Or to remember, where to find it.

Order-up-to Level (S) and Expected Back-Orders $B(S)$

$$B(S) = \mu - E\left(\frac{S - \mu}{\sigma}\right) = \sigma L\left(\frac{S - \mu}{\sigma}\right)$$

$$I(S) = S - \mu + B(S) = S - \mu + \sigma L\left(\frac{S - \mu}{\sigma}\right)$$

Medtronic Example: Distribution Center

- Consider 3 weeks lead time,
hence $l + 1 = 4$
- Mean demand during 4 periods = **322.4**
- Standard deviation of demand during 4 periods = **117.62**

S	F(S)	B(S)
400	0.745293	17.98095318
425	0.808478	12.42448422
450	0.861006	8.315740838
475	0.902753	5.384688138
500	0.934472	3.3696506
525	0.95751	2.035853396
550	0.973508	1.186478483
575	0.984127	0.666465753
600	0.990866	0.360568939
625	0.994954	0.187763369
650	0.997326	0.094057535
675	0.99864	0.045301241
700	0.999337	0.020967991
725	0.99969	0.009322889
750	0.999861	0.003980394
775	0.99994	0.001631302
800	0.999976	0.000641566

If you know pick, e.g., $S = 625$, $B(625) = 0.19$ and
 $I(S) = S - \mu + B(S) = 625 - 322.4 + .19 = 302.79$
 with a service level of $F(S) = 99.5\%$.

Medtronic Example: Sales Representatives

- Consider 1 day lead time,
hence $l + 1 = 2$
- Mean demand during 2 periods = **0,58**

S	F(S)	B(S)
0	0.559898	0.58
1	0.884639	0.139898
2	0.978814	0.024538
3	0.997021	0.003352
4	0.999662	0.000374
5	0.999968	0.0000351

If you know pick, e.g., $S = 3$, $B(3) = 0.0034$ and
 $I(S) = S - \mu + B(S) = 3 - 0,58 + .0034 = 2.42$
with a service level of $F(S) = 99.7\%$.

Choosing the Optimal Service Level

Optimal Service Level (to Minimize Costs)

$$SL = \frac{c_u}{c_u + c_o}$$

Example: Sales Representative

- Holding costs?
 - annual holding costs = 35% of product price
 - daily holding costs = $\frac{35\%}{360} = 0.000972 \cdot \text{product price}$
 - $h := 0.000972 \cdot \text{product price}$
- Stock out costs per item per day?
 - assume margin of 75%
 - assume 50% of margin is lost
 - $b := 0.75 \cdot 0.5 \cdot \text{product price} = 0.375 \cdot \text{product price}$

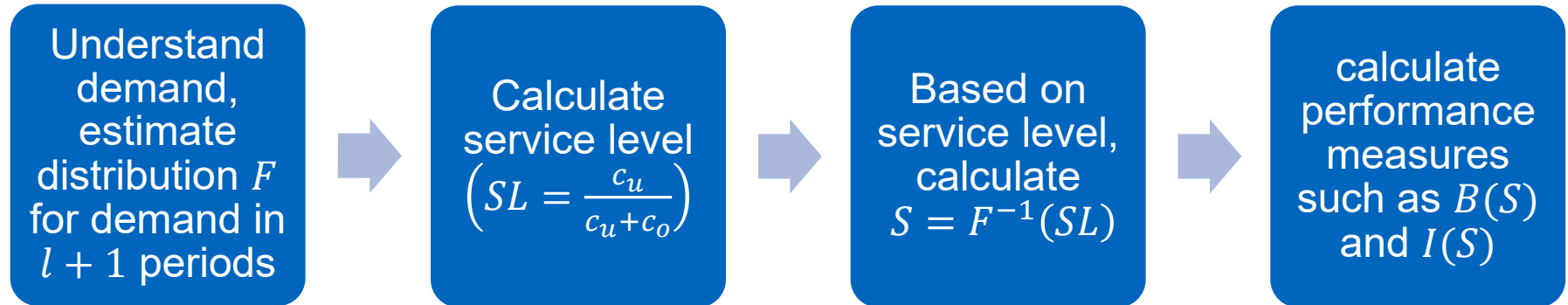
$$\Rightarrow SL = \frac{c_u}{c_u + c_o} = \frac{b}{b + h} = \frac{0.375}{0.375 + 0.000972} = 99.74\%$$

Example: Distribution center

- Holding costs?
- Stock out costs per item per day? How bad is a stock out really?
- Either use simulation or rule of thumb (e.g., 99% seems reasonable)

Bringing it all Together

Applying the Order-up-to Model



Differences between Order-up-to and Newsvendor Model

Time horizon

Overage costs/consequences

Lead time